

Speak math

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LEARN THE NOTATION. We will use math notation to design and specify our models. There are two benefits with this: it's explicit and easy to translate to code. Chapter 4 in the book goes through a first example of how to design a model, how to sample from it, and how to extend it.

We differ between two things in our models: outcome and predictor. An outcome is what we try to model. We want to predict the outcome. Often we use predictors in our models, i.e., things we believe influence the outcome (pp. 93 is key to this chapter). The chapter goes through a complete example where we do prior and posterior predictive checks and learn to plot things. Later in the chapter McElreath introduces polynomials (which we avoid) and splines (which we can use instead).

The simple linear regression model, a framework for estimating the association between a predictor variable and an outcome variable, is the basis we will stand on in this course.

SPURIOUS ASSOCIATIONS give rise to very many funky things. We use directed acyclic graphs to interpret these associations. These graphs comes in different patterns and depending on the patterns we get different conditional independencies, i.e., testable implications. We want to use this approach for multiple regression (i.e., having > 1 predictor). Carefully thinking about what variables to include as predictors will save us later down the road (i.e., included variable bias). The example in the book introduces > 1 predictors, so we can argue over which predictors should be included in a model.

By plotting things you will be able to see how well the model does its job. Learn how to do predictor residual plots (the average prediction error when we use all of the other predictor variables to model a predictor of interest), posterior prediction plots (check the model's implied predictions against the observed data), and counterfactual plots (displays the causal implications of the model). Each has its purpose.

THE CONCEPT OF MASKED RELATIONSHIPS is important to grasp. Basically it's about two predictors, where one influences the outcome positively and the other influences the outcome negatively.

We now also start using categorical variables, first by introduc-

Greek letters are used to show that it is a parameter we want to estimate. Yes, you will learn how to pronounce many Greek letters.

These two concepts will **not** be stressed in this course.

Spurious can be translated to 'fake'.

ing binary categories (e.g., true/false, female/male, 0/1), then we introduce many categories (> 2).

DIRECTED ACYCLIC GRAPHS (DAGs) are wonderful to use when we want to reason about causality. It provides us with testable implications, which we can use to test if our model ‘holds’.

In Chapter 6, a number of terms are introduced. *Non-identifiability* (the structure of the data and model does not make it possible to estimate a parameter), *collider bias* (adding a predictor induces statistical selection within the model), *multicollinearity* (strong association between two or more predictor variables, i.e., the multicollinear legs example), and *post-treatment bias* (the fungus example). The latter bias is the opposite of omitted variable bias, i.e., when we worry about a variable we don’t have access to; sometimes called included variable bias (i.e., we include variables to the model which *should not* be included).

In order to reason about the above terms we use DAGs. Four, and only four, elemental confounds exist in DAGs: fork, pipe, collider, and descendant. It’s important to know when we *should* and *should not* condition on a variable. i.e., include it in the model. Understand the *backdoor criterion*!

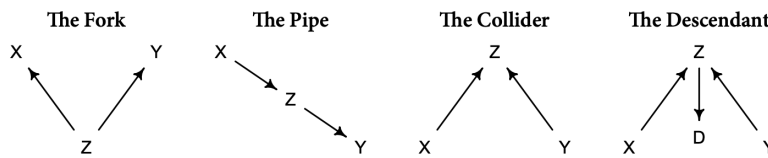


Figure 1: The four elemental confounds in DAGs.

And as always, relying on approach introduces assumptions that we need to keep in mind. The above rules (concerning DAGs) only hold if some important assumptions are met, which are listed below. (For details see Hernan and Robins (2024).)

- No spurious correlation: Correlation is not caused by random accident. The law of large numbers dictate that the more data we have, the more credible this assumption is.
- Consistency: The values of X (the treatment) you see are the actual values of X , or “the values of treatment under comparison correspond to well-defined interventions that, in turn, correspond to the versions of treatment in the data” (Hernan and Robins, 2024).
- Exchangeability: “the conditional probability of receiving every value of treatment, though not decided by the investigators, depends only on measured covariates” (Hernan and Robins, 2024).

- Positivity: in the observed data, there must be a non-zero probability of every individual or unit receiving any level of the treatment or exposure variable.
- Faithfulness: The causal effect does not vary over groups in a way that makes it average to 0 in the data. X does not have a positive effect 50% of the time and an identically powerful negative effect 50% of the time, which would average out to an effect of 0 in the population.

If either of these assumptions are not met, it can potentially break the relationship between the causal effect and observed data described by these rules.

Finally, a list of Greek characters and their pronunciation.

Character	Name	Character	Name
α	alpha <i>AL-fuh</i>	ν	nu <i>NEW</i>
β	beta <i>BAY-tuh</i>	ξ, Ξ	xi <i>KSIGH</i>
γ, Γ	gamma <i>GAM-muh</i>	$\omicron$	omicron <i>OM-uh-CRON</i>
δ, Δ	delta <i>DEL-tuh</i>	π, Π	pi <i>PIE</i>
ϵ	epsilon <i>EP-suh-lon</i>	ρ	rho <i>ROW</i>
ζ	zeta <i>ZAY-tuh</i>	σ, Σ	sigma <i>SIG-muh</i>
η	eta <i>AY-tuh</i>	τ	tau <i>TOW (as in cow)</i>
θ, Θ	theta <i>THAY-tuh</i>	υ, Υ	upsilon <i>OOP-suh-LON</i>
ι	iota <i>eye-OH-tuh</i>	ϕ, Φ	phi <i>FEE, or FI (as in hi)</i>
κ	kappa <i>KAP-uh</i>	χ	chi <i>KI (as in hi)</i>
λ, Λ	lambda <i>LAM-duh</i>	ψ, Ψ	psi <i>SIGH, or PSIGH</i>
μ	mu <i>MEW</i>	ω, Ω	omega <i>oh-MAY-guh</i>

References

M. Hernan and J. Robins. *Causal inference: What if*. Chapman & Hall/CRC Monographs on Statistics & Applied Probability. CRC Press, 2024. ISBN 9781420076165.